

Finite-Alphabet Plug-In Convergence for Shared-Exclusions PID

Exact-real theorem, local moduli, an anytime i.i.d. envelope,
and a formal evidence boundary

pid-rs project analysis

22 July 2026

Abstract

This document gives new project-defined validation for existing information measures in `pid-rs`. Its primary object is the Wibral-lineage categorical shared-exclusions partial information decomposition (SxPID). The paper-defined functional is not new. The document proves exact-real plug-in convergence on fixed finite alphabets, gives explicit local continuity moduli, and composes them with a conservative time-uniform i.i.d. concentration event. A secondary corollary covers the Williams–Beer $I_{\min}$ functional and fixed Shannon combinations. A frozen-transform corollary states the exact conditions under which an independently trained finite map preserves the argument. The theorem does not prove convergence of binary64 program outputs, dependent-window validity, drift calibration, or scientific novelty. Invalid stronger claims and counterexamples remain in the document.

A separate repository paper on dependency-colored SxPID gives a categorical result for a declared deterministic coloring. Its common-law theorem requires all complete rows to share one finite law. Complete rows in each color must be mutually independent. It also separates nonidentical-law drift. That result does not change the i.i.d. and ergodic claims in this paper, and it does not cover generic mixing, adaptive colors, or continuous estimators.

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1 Status and provenance

The mathematical object and the validation artifact have different origins.

Object	Origin	Result in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its part-whole lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [7, 4]	Exact-real convergence for two to four sources, local moduli, and bounded code evidence
$I_{\min}$	Paper-defined by Williams and Beer [11]	Secondary finite-minimum corollary for two and three sources
Shannon entropy and mutual information	Paper-defined classical quantities [10]	Exact-real convergence and entropy continuity bounds
Discrete co-information and O-information	Paper-defined quantities [2, 8, 9]	Convergence as fixed finite entropy combinations
Target-free normalized ratios	Project-defined analogues in <code>pid-rs</code>	Positive-denominator ratio convergence
Target-conditioned normalized ratios	Paper-defined ratios with a project-defined report policy [5]	Positive-denominator ratio convergence and a status boundary
Fitted quantizer plus categorical PID	Project-defined composition; quantization has published background [3]	Conditional frozen-transform corollary

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. This document defines no new estimator or public API. The exact Rust routes covered by the result are:

- SxPID namespace `stable::categorical`: `discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`, `discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`;
- $I_{\min}$ namespace `stable::imin`: `imin_pid2` and `imin_pid3`;
- Shannon namespace `diagnostics`: `entropy_discrete`, `joint_entropy_discrete`, `co_information_pairwise_discrete`, and `o_information_discrete`;
- normalized-report namespace `diagnostics`: `red_degree_discrete`, `vul_degree_discrete`, `average_degree_of_redundancy`, and `average_degree_of_vulnerability`;
- fitted SxPID namespace `stable::quantized`: `fitted_quantized_sxpid2`, `fitted_quantized_sxpid3`, and `fitted_quantized_sxpid_n`;
- fitted $I_{\min}$ namespace `stable::imin`: `imin_pid2_quantized` and `imin_pid3_quantized`.

2 Finite setting and shared-exclusions quantities

Fix a source count m , finite source alphabets $\mathcal{S}_1, \dots, \mathcal{S}_m$, a finite target alphabet $\mathcal{T}$, and the joint alphabet

$$\mathcal{Z} = \mathcal{S}_1 \times \dots \times \mathcal{S}_m \times \mathcal{T}. \quad (1)$$

Let P be one probability law on $\mathcal{Z}$, and put

$$S = \text{supp}(P), \quad p_{\min} = \min_{z \in S} P(z) > 0. \quad (2)$$

All logarithms are natural, so all information values use nats. The redundancy lattice is fixed and finite. Its nodes are nonempty antichains of nonempty source subsets. In **pid-rs**, $m \in \{2, 3, 4\}$ for SxPID and $m \in \{2, 3\}$ for $I_{\min}$.

For observations $Z_1, Z_2, \dots$, define the prefix empirical law

$$\hat{P}_n(z) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{Z_j = z\}. \quad (3)$$

Fix a supported realization $z = (s_1, \dots, s_m, t)$ and a lattice node α . Define

$$A_\alpha(s) = \bigcup_{a \in \alpha} \{S_a = s_a\}, \quad (4)$$

$$B_\alpha(s, t) = A_\alpha(s) \cap \{T = t\}, \quad (5)$$

$$C(t) = \{T = t\}. \quad (6)$$

Each of these three events contains z . Its P -mass is therefore at least $P(z) \geq p_{\min}$. For any law Q whose required masses are positive, the cumulative informative and misinformative terms are

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha(s)), \quad (7)$$

$$c_\alpha^-(z; Q) = \log \frac{Q(C(t))}{Q(B_\alpha(s, t))}. \quad (8)$$

The cumulative net term is $c_\alpha^{\text{sx}} = c_\alpha^+ - c_\alpha^-$. A fixed Möbius matrix M maps each cumulative vector to its pointwise atom vector. The averaged atom under Q is the Q -weighted sum over joint realizations.

3 Deterministic convergence and sampling corollary

Theorem 3.1 (Deterministic plug-in implication). *Let Q_n be probability laws on $\mathcal{Z}$. Assume*

(i) $Q_n(z) \rightarrow P(z)$ for every $z \in \mathcal{Z}$; and

(ii) $\text{supp}(Q_n) \subseteq S$ for every n .

Then there is a finite N_0 such that $\text{supp}(Q_n) = S$ for every $n \geq N_0$. On this tail, all supported-event denominators are positive. The following exact-real quantities converge to their corresponding values under P :

1. every supported-realization SxPID cumulative informative and misinformative term;
2. every informative, misinformative, and signed-net Möbius atom;
3. every probability-weighted averaged SxPID atom;

4. every supported-target $I_{\min}$ specific-information value and every fixed two- or three-source $I_{\min}$ atom; and
5. every fixed finite-alphabet Shannon entropy, mutual information, co-information, and O -information expression formed from the same law.

Pointwise SxPID convergence is keyed by the joint realization, not by a returned-vector index.

Proof. For each $z \in S$, coordinate convergence and $P(z) > 0$ give $Q_n(z) > P(z)/2$ for all sufficiently large n . The support is finite, so one finite N_0 works for every supported cell. Assumption (ii) excludes new cells. This proves support equality on the tail.

Fix z and α . Event masses are finite linear sums of cell masses, so the masses of A_α , B_α , and C converge. Their limits are at least $P(z) > 0$. Continuity of the logarithm on $(0, \infty)$ gives convergence of c_α^+ and c_α^- . The fixed finite linear map M preserves convergence. Subtraction gives the net atoms. After support stabilization, an average is a fixed finite sum whose weights and pointwise values both converge.

For a nonempty source subset a and a supported target value t , write

$$J_a(t; Q) = \sum_{s_a: Q(s_a, t) > 0} \frac{Q(s_a, t)}{Q(t)} \log \frac{Q(s_a, t)}{Q(s_a)Q(t)}. \quad (9)$$

After support stabilization, every positive marginal cell and required denominator is fixed and positive. Each finite term is continuous. A finite minimum of continuous functions is continuous, including at a tie. Target averaging and the fixed Möbius map preserve convergence. A tie can remove differentiability, but it does not remove continuity.

Finally, define $h(0) = 0$ and $h(x) = -x \log x$ for $x > 0$. The function h is continuous on $[0, 1]$. Each finite-alphabet entropy is a finite sum of h . Marginalization and fixed finite linear combinations preserve convergence. $\square$

Corollary 3.2 (I.i.d. and stationary ergodic sampling). *The assumptions of Theorem 3.1 hold almost surely when either*

- (i) *the observations are i.i.d. with law P ; or*
- (ii) *the process is strictly stationary and ergodic with one-time marginal law P .*

Strict stationarity without ergodicity is not sufficient.

Proof. Apply the strong law, or the pointwise ergodic theorem, to the indicator of each cell. A finite intersection gives simultaneous coordinate convergence. If $P(z) = 0$, every time index has zero probability of that cell. A countable union shows that no such cell occurs, almost surely. Thus, the empirical support is always a subset of S on one probability-one event. $\square$

4 Quantitative local moduli for SxPID

Let p be the probability vector of P , and let q be another probability law on the same finite joint alphabet. Define

$$\delta = \|q - p\|_1, \quad L = \log(1/p_{\min}), \quad (10)$$

and assume $\delta \leq p_{\min}/2$. For every event E ,

$$|q(E) - p(E)| \leq \delta. \quad (11)$$

This deliberately loose inequality is sufficient for the following stated constants.

Theorem 4.1 (SxPID local bounds). *For each supported realization and lattice node,*

$$|c_\alpha^+(z; q) - c_\alpha^+(z; p)| \leq \frac{2\delta}{p_{\min}}, \quad (12)$$

$$|c_\alpha^-(z; q) - c_\alpha^-(z; p)| \leq \frac{4\delta}{p_{\min}}. \quad (13)$$

Let

$$\|M\|_\infty = \max_i \sum_j |M_{ij}|. \quad (14)$$

The pointwise atom errors obey

$$\text{informative: } |\Delta\pi^+| \leq \frac{2\|M\|_\infty \delta}{p_{\min}}, \quad (15)$$

$$\text{misinformative: } |\Delta\pi^-| \leq \frac{4\|M\|_\infty \delta}{p_{\min}}, \quad (16)$$

$$\text{net: } |\Delta\pi^{\text{sx}}| \leq \frac{6\|M\|_\infty \delta}{p_{\min}}. \quad (17)$$

If also $\text{supp}(q) \subseteq S$, the averaged errors obey

$$\text{informative: } |\Delta\bar{\pi}^+| \leq \|M\|_\infty \left(\frac{2}{p_{\min}} + L \right) \delta, \quad (18)$$

$$\text{misinformative: } |\Delta\bar{\pi}^-| \leq \|M\|_\infty \left(\frac{4}{p_{\min}} + L \right) \delta, \quad (19)$$

$$\text{net: } |\Delta\bar{\pi}^{\text{sx}}| \leq \|M\|_\infty \left(\frac{6}{p_{\min}} + L \right) \delta. \quad (20)$$

Proof. Each required event E has $p(E) \geq p_{\min}$. Equations (11) and $\delta \leq p_{\min}/2$ give $q(E) \geq p_{\min}/2$. The mean-value theorem for log gives

$$|\log q(E) - \log p(E)| \leq \frac{|q(E) - p(E)|}{\min\{q(E), p(E)\}} \leq \frac{2\delta}{p_{\min}}. \quad (21)$$

The informative term contains one logarithm. The misinformative term contains two. The row-sum norm of M gives the first two atom bounds in (15); the triangle inequality gives the net bound.

For one atom component π , decompose its averaged error as

$$\left| \sum_{z \in S} q(z) \pi(z; q) - \sum_{z \in S} p(z) \pi(z; p) \right| \leq \sum_{z \in S} q(z) |\pi(z; q) - \pi(z; p)| \quad (22)$$

$$+ \sum_{z \in S} |q(z) - p(z)| |\pi(z; p)|. \quad (23)$$

At p , each cumulative informative and misinformative coordinate lies in $[0, L]$. For the net term, $c_\alpha^+ - c_\alpha^- \in [-L, L]$. Apply the row-sum norm once. This gives one factor of L for each of the three weight-change terms and proves the averaged bounds. $\square$

Remark 4.2 (Why the common-support condition matters). Under $\delta \leq p_{\min}/2$, the condition $\text{supp}(q) \subseteq S$ implies $\text{supp}(q) = S$. Without common support, a new cell of mass ε can contribute at order $\varepsilon \log(1/\varepsilon)$. No fixed constant C can bound this by $C\varepsilon$ as $\varepsilon \downarrow 0$. Thus, the support condition cannot be removed from a uniform linear guarantee of this form.

These are exact-real law-level inequalities. They do not include binary64 rounding, platform logarithm variation, resource rejection, or implementation refinement. The Rust SxPID and $I_{\min}$ paths reject sample counts above 2^{53} .

5 Secondary $I_{\min}$ modulus

This section is a secondary validation control. It is not the research priority of the document. Assume $\text{supp}(q) \subseteq S$ and $\delta \leq p_{\min}/2$. For a supported target t , let

$$r = q(s_a, t), \quad u = q(s_a), \quad v = q(t), \quad w = \frac{r}{v}, \quad \ell = \log \frac{r}{uv}. \quad (24)$$

Marginalization contracts joint L^1 distance. Every positive p -marginal is at least $p_{\min}$, and every corresponding q -marginal is at least $p_{\min}/2$. Hence

$$\sum_{s_a} |w_q(s_a) - w_p(s_a)| \leq \frac{4\delta}{p_{\min}}, \quad (25)$$

and

$$|\ell_q - \ell_p| \leq \frac{6\delta}{p_{\min}}. \quad (26)$$

To see (25), put $r_q(s_a) = q(s_a, t)$, $v_q = q(t)$, and write

$$\sum_{s_a} \left| \frac{r_q(s_a)}{v_q} - \frac{r_p(s_a)}{v_p} \right| \leq \frac{\sum_{s_a} |r_q(s_a) - r_p(s_a)|}{v_q} + \frac{|v_q - v_p|}{v_q} \quad (27)$$

$$\leq \frac{2\delta}{p_{\min}/2}. \quad (28)$$

Each of the three logarithm factors changes by at most $2\delta/p_{\min}$. Equation (26) is their sum. Also, $|\ell_p| \leq \log(2/p_{\min})$ under the stated neighborhood. Therefore,

$$|J_a(t; q) - J_a(t; p)| \leq \left[4 \log \left(\frac{2}{p_{\min}} \right) + 6 \right] \frac{\delta}{p_{\min}}. \quad (29)$$

A finite minimum is one-Lipschitz in the sup norm. If C_J is the coefficient of δ in (29), define

$$R_\alpha(t; Q) = \min_{a \in \alpha} J_a(t; Q), \quad (30)$$

$$I_{\cap}^{\min}(\alpha; Q) = \sum_t Q(t) R_\alpha(t; Q). \quad (31)$$

Each $J_a(t; p)$ is a Kullback–Leibler divergence, so it is nonnegative. Also,

$$J_a(t; p) = \sum_{s_a} p(s_a | t) \log \frac{p(t | s_a)}{p(t)} \leq -\log p(t) \leq L. \quad (32)$$

Thus, $0 \leq R_\alpha(t; p) \leq L$. Marginalization gives $\|q_T - p_T\|_1 \leq \delta$. Therefore,

$$\left| I_{\cap}^{\min}(\alpha; q) - I_{\cap}^{\min}(\alpha; p) \right| \leq \sum_t q(t) |R_\alpha(t; q) - R_\alpha(t; p)| \quad (33)$$

$$+ \sum_t |q(t) - p(t)| R_\alpha(t; p) \quad (34)$$

$$\leq (C_J + L)\delta. \quad (35)$$

Since $L = \log(1/p_{\min})$, the last line is the target-averaged redundancy bound. The fixed $I_{\min}$ Möbius matrix multiplies this bound by its absolute row-sum norm. A minimum tie preserves continuity but can remove differentiability and a generic normal limit.

6 Shannon and ratio bounds

Let one entropy term have alphabet size $K \geq 2$, and put

$$\varepsilon = \frac{1}{2} \|q - p\|_1. \quad (36)$$

Apply Audenaert's bound [1] to diagonal density matrices with diagonals p and q . Their trace distance is ε , and their von Neumann entropies are $H(p)$ and $H(q)$. For $0 \leq \varepsilon \leq 1 - 1/K$, the Fannes–Audenaert bound gives

$$|H(q) - H(p)| \leq g_K(\varepsilon) := \varepsilon \log(K - 1) + h_2(\varepsilon), \quad (37)$$

where

$$h_2(x) = -x \log x - (1 - x) \log(1 - x), \quad h_2(0) = h_2(1) = 0. \quad (38)$$

The function g_K is increasing only up to $1 - 1/K$. For a marginal A with alphabet size $K_A \geq 2$ and only a joint variation bound available, the safe bound is

$$|H(q_A) - H(p_A)| \leq g_{K_A}(\min\{\varepsilon, 1 - 1/K_A\}). \quad (39)$$

If $K_A = 1$, the entropy difference is zero. A fixed entropy linear combination is bounded by the sum of the absolute coefficients times the term-specific bounds.

Lemma 6.1 (Ratio perturbation). *Let $D_p \geq d > 0$, $|D_q - D_p| \leq d/2$, and $|N_p| \leq B$. Then*

$$\left| \frac{N_q}{D_q} - \frac{N_p}{D_p} \right| \leq \frac{2|N_q - N_p|}{d} + \frac{2B|D_q - D_p|}{d^2}. \quad (40)$$

Proof. The denominator condition gives $D_q \geq d/2$. Decompose the difference into $(N_q - N_p)/D_q + N_p(D_p - D_q)/(D_q D_p)$ and apply the triangle inequality. $\square$

Target-free normalized limits require positive population joint entropy. Target-conditioned normalized limits require positive population joint mutual information. Every target-conditioned input term must describe the same law, target, source order, units, preprocessing, evaluation sample, and estimand. For the convergence claim, each input sequence must be the corresponding exact-real finite-alphabet plug-in term formed from the same empirical law. A positive denominator alone does not establish this coherence.

For a report threshold τ , a population denominator strictly above τ guarantees eventual **Defined** status. A denominator below τ guarantees an eventual status other than **Defined**. Equality gives no eventual status guarantee.

7 A time-uniform i.i.d. envelope

This section uses Hoeffding's published inequality [6] and union bounds. Their composition here is project-defined validation and makes no scientific-novelty claim.

Theorem 7.1 (All-prefix empirical-law bound). *Assume i.i.d. sampling from one fixed K -cell law. Let $0 < \alpha < 1$ and*

$$\alpha_n = \frac{6\alpha}{\pi^2 n^2}. \quad (41)$$

With probability at least $1 - \alpha$, simultaneously for every $n \geq 1$,

$$\|\hat{P}_n - P\|_1 \leq B_n := K \sqrt{\frac{\log(K\pi^2 n^2/(3\alpha))}{2n}}. \quad (42)$$

Proof. For one cell, Hoeffding's inequality gives

$$\mathbb{P}(|\hat{P}_n(z) - P(z)| \geq x) \leq 2e^{-2nx^2}. \quad (43)$$

Set

$$x_n = \sqrt{\frac{\log(K\pi^2 n^2 / (3\alpha))}{2n}}. \quad (44)$$

The failure probability for one cell at time n is at most α_n/K . A union bound over K cells and all n costs

$$\sum_{n \geq 1} \alpha_n = \frac{6\alpha}{\pi^2} \sum_{n \geq 1} \frac{1}{n^2} = \alpha. \quad (45)$$

On the complement, summing the K coordinate bounds gives (42). $\square$

Define

$$N_0 = \min \{r \geq 1 : B_n \leq p_{\min}/2 \text{ for every } n \geq r\}. \quad (46)$$

The integer N_0 is finite because $B_n \rightarrow 0$. Intersect the concentration event with the probability-one support-containment event of Corollary 3.2. On the intersection, support equality holds for all $n \geq N_0$. Substitute B_n for the actual δ in Theorem 4.1.

For a fixed prefix N , a separate support-discovery bound is

$$\mathbb{P}(\text{supp}(\hat{P}_N) \neq S) \leq |S|e^{-Np_{\min}} \leq Ke^{-Np_{\min}}. \quad (47)$$

A usable N_0 needs a known positive lower bound on $p_{\min}$. An empirical minimum cannot replace it because an unobserved rare state can have positive population mass. The simultaneous L^1 bound can be read at an almost-surely finite data-dependent index. Support and local-bound conclusions additionally need that index to be at least N_0 .

The theorem controls an exact-real empirical law. It is not a confidence sequence for Rust output. It does not include floating-point error, the platform logarithm, resource failure, dependence, drift, rejection selection, or the 2^{53} sample-count limit.

8 Independent frozen transforms

Let $\mathcal{G}$ be the sigma-field generated by a training artifact and its fitted transform. Assume that the training artifact is independent of the raw evaluation sequence. Let the raw evaluation space $\mathcal{X}$ be standard Borel. Let $\mathcal{Z}$ be one fixed finite output alphabet, and let $\perp \notin \mathcal{Z}$ be a failure symbol. Assume a map

$$Q : \Omega \times \mathcal{X} \longrightarrow \mathcal{Z} \cup \{\perp\} \quad (48)$$

is measurable from $(\Omega \times \mathcal{X}, \mathcal{G} \otimes \mathcal{B}(\mathcal{X}))$ to $(\mathcal{Z} \cup \{\perp\}, 2^{\mathcal{Z} \cup \{\perp\}})$. The output block structure is fixed and finite. The map is frozen for all rows and prefixes. Evaluation rows W_j are conditionally i.i.d. given $\mathcal{G}$. Assume

$$\mathbb{P}(Q(W_1) \neq \perp \mid \mathcal{G}) = 1 \quad \text{almost surely.} \quad (49)$$

Corollary 8.1 (Conditional push-forward target). *For almost every training outcome, the transformed rows are conditionally i.i.d. with random push-forward law*

$$P_Q(z) = \mathbb{P}(Q(W_1) = z \mid \mathcal{G}), \quad z \in \mathcal{Z}. \quad (50)$$

The conclusions of Corollary 3.2 apply to this law. The limit is the PID or Shannon quantity of P_Q . In general, it is not the same as the quantity of the unconditional mixture.

If an error-policy transform violates the zero-failure assumption, let $\eta = \eta(\mathcal{G})$ be its conditional per-row failure probability. For a training outcome with $\eta > 0$, the first n rows all succeed with probability $(1 - \eta)^n$, and the infinite prefix fails with conditional probability one. Unconditionally, this failure mechanism has probability $\mathbb{P}(\eta > 0)$. It has probability one only when $\eta > 0$ almost surely.

Conditioning a complete prefix on success does not preserve the original-law claim. Explicit rejection sampling can produce a sequence that is conditionally i.i.d. given $\mathcal{G}$ from a conditional law when success has positive probability. That sequence has a different estimand and needs a declared sampling contract. Clamping supplies a total range map for otherwise valid finite inputs, but it defines a tail-clamped categorical estimand.

Same-row fitting, changing transforms, arbitrary pooling across different fitted maps, and target-adaptive fitting on evaluation rows are outside the corollary.

9 Counterexamples and invalid stronger claims

Lens	Construction	Consequence
Support face	Add a new cell of mass $\varepsilon \downarrow 0$.	A local informative term can grow as $-\log \varepsilon$; uniform linear averaged bounds fail without common support.
Probability	Draw $C \sim \text{Bernoulli}(1/2)$ and set $Z_j = C$ for every j .	The process is stationary but not ergodic; a path sees only one state.
Rare state	Give one state mass ε .	It is absent after n rows with probability $(1 - \varepsilon)^n$; no uniform deterministic discovery time exists.
$I_{\min}$ tie	Cross a law where two specific-information values exchange order.	The minimum is continuous but need not be differentiable; a generic delta-method normal limit does not follow.
Transform	Fit finite min/max ranges, then evaluate an independent standard-normal held-out law with an error policy.	Every finite range leaves positive out-of-range mass; infinite-prefix success has probability zero for each positive-failure training outcome.
Mixture	Randomly choose a correlated or anticorrelated binary push-forward law.	Each conditional law has mutual information $\log 2$, but their equal mixture is independent and has mutual information zero.
Indexing	Append a rare realization that is lexicographically first.	Returned vector positions shift; pointwise convergence requires realization keys.
Numerics	Exceed exact binary64 integer counts or change the platform logarithm.	The exact-real theorem does not become a portable executable theorem.

Lens	Construction	Consequence
Downstream	Reuse overlapping windows, drift the law, or select complete samples after inspecting them.	The i.i.d. proof and its envelope do not apply.

The following stronger claims are invalid and are retained for audit:

1. strict stationarity is sufficient without ergodicity;
2. all-unique samples identify population support;
3. pointwise atoms converge by returned-vector index;
4. one deterministic support-discovery time works for every finite law;
5. complete-success conditioning preserves the original error-policy estimand;
6. the conditional fitted-transform limit equals the PID of the unconditional mixture;
7. foldwise PMFs or atoms can be pooled when each fold uses a different map;
8. the Lean artifact proves the stochastic theorem or the Rust implementation;
9. a fixture tolerance is a portable binary64 error theorem;
10. unrestricted exact sign and tie evaluation is always practical after denominator clearing;
11. the all-prefix i.i.d. bound covers sliding or dependent windows; and
12. continuity at an $I_{\min}$ tie implies a normal limit.

10 Formal and executable evidence boundary

The Lean 4.32 artifact proves only these deterministic exact-real lemmas:

- eventual positivity at a positive limit;
- convergence of a fixed finite event mass;
- positivity of a finite sum of nonnegative cell masses when the event contains a positive cell;
- sequential-limit preservation by $\log$ and $-\log$ at positive limits, and by log ratios at positive limits;
- sequential-limit preservation by $-x \log x$, including at zero;
- preservation of convergence by fixed finite linear and weighted sums;
- preservation of convergence by a finite nonempty minimum, including ties; and
- preservation by a finite weighted sum of finite positive-limit log-ratio combinations.

It does not encode empirical probability laws, stochastic limit theorems, Sx events, the redundancy lattice, $I_{\min}$, or Rust refinement. The checker binds the complete Lake manifest and all nine package revisions. It verifies each dependency checkout’s root, revision, origin, and clean status with global and system Git configuration and Git environment routing disabled. It retains the checkout’s local configuration to verify the recorded origin. The checked project sources cannot contain the tokens `admit`, `axiom`, `constant`, `sorry`, or `sorryAx`. The checker builds the project and replays its declarations with Lean’s bundled kernel checker. CI runs the same build and kernel checks.

The independent 100-digit Decimal fixture covers all coordinates in listed two-, three-, and four-source SxPID tables, listed two- and three-source $I_{\min}$ tables, minimum-tie crossings, realization-key changes, and pointwise omission of an absent realization on the listed support face. The Rust test separately checks bit-identical outputs from fitted-quantizer wrappers and direct calls on the same categorical matrices. The committed Decimal comparisons use the absolute envelope

$$64\text{ f64}::\text{EPSILON nats.} \quad (51)$$

This is bounded implementation evidence. It is not the convergence proof, external review, population validation, or a global floating-point bound.

11 Downstream contract boundary

Consumer	Covered path	Uncovered path
Prisoma	Use conditionally i.i.d. cases or episodes from one common conditional law. Use one evaluation row per unit, or separately establish the row law. Fit and freeze transforms on an independent partition.	Independent non-identical units, dependent rows within an episode, same-row fitting, pooling outputs from different maps, or other dependence without a theorem.
Galadriel	Use direct categorical SxPID on i.i.d. rows with a fixed alphabet.	Continuous KSG on dependent temporal windows, same-window scaling, added-noise studies, circular delete-block work, or complete-sample rejection after a degeneracy check.
Haldir	Use i.i.d. or strictly stationary and ergodic fixed-law rows for finite-alphabet entropy, co-information, O-information, SxPID, and positive-denominator ratio limits.	A fixed one-time marginal without ergodicity, drift, CUSUM, alarms, overlapping windows, post-alarm estimation, or calibration.
Crebain	Use stable categorical and fitted-transform contracts after an explicit dependency and input-contract review.	No compatibility or adapter claim follows from this theorem.

No downstream repository is a runtime dependency of `pid-rs`. This audit states mathematical preconditions. It does not claim downstream qualification.

12 Reproduction

Run these commands from the repository root:

```
python3 scripts/generate-finite-alphabet-plugin-oracle.py
cargo test --locked -p pid-core --test finite_alphabet_plugin_oracle
python3 scripts/check-lean-finite-convergence.py
scripts/check-finite-alphabet-convergence-pdf.sh
python3 scripts/check-method-catalog.py
```

13 Correction ledger

The review process corrected the following defects in the current reviewed draft:

1. A local-bound draft reused $p_{\min}$ from a law different from the local reference law. The final statement sets $p = P$.
2. A Shannon draft substituted joint variation into a nonmonotone marginal Fannes–Audenaert envelope. Equation (39) now clips at $1 - 1/K_A$.
3. A transform draft did not require the random map to be training-measurable. A map such as $Q(\omega, x) = x \text{ xor } W_1(\omega)$ can otherwise destroy conditional independence.
4. A theorem draft omitted the support-stabilized tail and allowed an empty antichain.
5. A stopping statement omitted the condition that support conclusions need an index at least N_0 .
6. A failure statement treated a random η as positive almost surely. The correct unconditional probability is $\mathbb{P}(\eta > 0)$.
7. A rejection statement did not distinguish complete-prefix conditioning from a newly declared rejection-sampling estimand.
8. A status statement incorrectly made strict threshold separation necessary for every eventual **Defined** sequence. At threshold zero, the nested count path $(n_{00}, n_{01}, n_{10}, n_{11}) = (2, 0, 0, 2)$ followed by repeated additions 01, 10, 00, 11 converges to independence while empirical mutual information stays positive.
9. A downstream draft omitted common-law and ergodicity conditions.
10. An evidence draft attributed quantizer inputs to the Decimal JSON. They occur only in the companion Rust test.
11. A valid first averaged-net bound used $2L$. Applying the net cumulative range directly gives the tighter L term in Theorem 4.1.
12. An intermediate entropy draft asserted $p_{\min} \leq 1/K$. This is false when K is the declared alphabet size and P has sparse support; one-cell support has $p_{\min} = 1$. No accepted bound uses this claim. Alphabet size and supported-cell mass remain separate.

14 Open Wibral-lineage work

The next research priority is shared-exclusions PID, not an extension of $I_{\min}$. Open work is:

1. a dependence-aware SxPID result for generic mixing or temporal processes that do not admit the explicit independence coloring in the separate dependency-colored result;
2. a sequential contract that separates fixed-law monitoring from drift and post-alarm estimation;
3. a triangular-array or cross-fit theorem for changing fitted maps;
4. practical certified sign and tie evaluation for empirical Sx expressions;
5. a deductive refinement to binary64 or interval arithmetic;
6. a Lean encoding of the empirical-law step, Sx events, and the Sx redundancy lattice; and
7. mathematically valid multivariate and mixed-dimensional extensions needed by ecosystem consumers.

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